RK3568 NVR DEMO Board User Guide

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Release Version: V1.3
Release Date: 2021.06.24

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Preface

Overview

This document is intended to introduce the basic functions and hardware characteristics, multi-function hardware configurations and software debug methods of RK3568 NVR DEMO board, aiming to help developers to get start with RK3568 NVR DEMO board and NVR solution more quickly and correctly.

Product version

The product versions corresponding to this document are as follows:

Product name	Product version
RK3568 NVR DEMO	RK_NVR_DEMO_RK3568_DDR4P216SD4_V12_20201217

Intended Audience

This document (this guide) is mainly intended for：

- Technical support engineers
- Single board hardware development engineers
- Embedded software development engineers
- Test engineers

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Revision History

This revision history recorded description of each version.

Version No.	Author	Revision Date	Revision History
V1.0	WHB	2020-12-24	Initial Release
V1.2	WHB	2021-02-05	Modify debug port according to NVR DEMO board
V1.3	WHB	2021-06-24	Detailed description of optimization

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Acronyms

These Acronyms includes the abbreviations of commonly used phrases in this document.

Acronym	Description in English	Description in Chinese
CPU	Central Processing Unit	中央处理器
NPU	Neural Network Processing Unit	神经网络处理器
VPU	Video Processing Unit	视频处理器
DDR	Double Data Rate	双倍速率同步动态随机存储器
eMMC	Embedded Multi Media Card	内嵌式多媒体存储卡
eDP	Embedded Display Port	嵌入式数码音视讯传输接口
HDMI	High Definition Multimedia Interface	高清晰度多媒体接口
I2C	Inter-Integrated Circuit	内部整合电路(两线式串行通讯总线)
LDO	Low Drop Out Linear Regulator	低压差线性稳压器
DCDC	Direct Current to Direct Current	直流电转直流电
SARADC	Successive Approximation Register Analog to Digital Converter	逐次逼近寄存器型模数转换器
UART	Universal Asynchronous Receiver/ Transmitter	通用异步收发传输器
PWM	Pulse Width Modulation	脉冲宽度调制
MIPI	Mobile Industry Processor Interface	移动产业处理器接口
LVDS	Low-Voltage Differential Signaling	低电压差分信号
PMIC	Power Management IC	电源管理芯片
PMU	Power Management Unit	电源管理单元
RK/Rockchip	Rockchip Electronics Co.,Ltd.	瑞芯微电子股份有限公司
USB	Universal Serial Bus	通用串行总线
SATA	Serial Advanced Technology Attachment	串行高级技术附件
PCIe	Peripheral Component Interconnect Express	外围组件快速互连
RGB	Red,Green,Blue；RGB color mode is a color standard in industry	红绿蓝，RGB色彩模式，是工业界的一种颜色标准
VGA	Video Graphics Array	电脑显示视频图像标准接口
ATX	Advanced Technology Extended	高级技术扩展电源
ADB	Android Debug Bridge	安卓调试桥
IR	Infrared Radiation	红外线
RGMII	Reduced Gigabit Media Independent Interface	精简吉比特介质独立接口

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Contents

Contents ..... **V**
Figures ..... **VII**
Tables ..... **VIII**
1 Overview **1**
1.1 RK3568 Introduction 1
1.2 RK3568 Block Diagram 1
1.3 System Introduction 2
1.3.1 System Diagram 2
1.3.2 Functions Overview 2
1.3.3 Functional Interfaces 3
1.3.4 Functional Modules Layout 4
1.4 Components 5
1.5 Power on and off 5
1.6 Driver Upgrade 6
1.6.1 USB Driver Installation 6
1.6.2 Driver Upgrade 6
1.7 Serial Port Debugging 8
1.7.1 Serial Port Debugging Tool 8
1.7.2 ADB Debug 9
2 Hardware Introduction **11**
2.1 The Picture of NVR DEMO Board 11
2.2 Power Diagram 12
2.3 I2C Address 12
2.4 Extension Connectors 13
2.5 Reference Materials 14
3 Modules Introduction **15**
3.1 Power Input 15
3.2 Memory 17
3.3 RTC Circuit 18
3.4 Keys Input 18
3.5 Infrared Receiver 19
3.6 UART Interface 19
3.7 UART Debug Interface 20
3.8 RS485 and RS232 Interface 20
3.9 MIPI/LVDS Output Interface 21
3.10 HDMI Output Interface 22
3.11 VGA Output Interface 23
3.12 Audio Input and Output Interface 23
3.13 USB OTG/HOST Interface 24
3.14 Ethernet Interface 25
3.15 Mini PCIe Interface 26
3.16 SATA Interface 27

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3.17 Indicator Lights 28
4 Notice 29
4.1 Notice 29

Figures

Figure 1-1 RK3568 Block Diagram 1
Figure 1-2 RK3568 NVR DEMO System Diagram 2
Figure 1-3 NVR DEMO Board Functional Interfaces Layout (TOP Surface) 4
Figure 1-4 NVR DEMO Board Functional Interfaces Layout (Bottom Surface) 5
Figure 1-5 Driver Installed Successfully 6
Figure 1-6 Loader Flashing Mode 7
Figure 1-7 Maskrom Flashing Mode 8
Figure 1-8 Current COM Number 8
Figure 1-9 Serial Port Tool Configuration Interface 9
Figure 1-10 Serial Port Tool Debugging Interface 9
Figure 1-11 ADB Connected Successfully 10
Figure 2-1 The Picture of NVR DEMO Board 11
Figure 2-2 RK3568 NVR DEMO Board Power Diagram 12
Figure 2-3 Vertical Double Row 30 PIN PCB Package with 1mm Pitch 14
Figure 3-1 DC12V Input and Front-end Buck Converter 15
Figure 3-2 ATX Main POWER and ATX +12V DC Interface 16
Figure 3-3 20pin ATX Main POWER Version1.0 Interface Signal 16
Figure 3-4 4pin ATX +12V DC Interface Signal 16
Figure 3-5 DDR4 and eMMC 17
Figure 3-6 The Location theTest Point When Using eMMC Entering Maskrom Flashing Mode 17
Figure 3-7 The Location of SPI Nand and Test Point When Using SPI Nand Entering Maskrom 18
Figure 3-8 RTC Circuit 18
Figure 3-9 Keys Location 19
Figure 3-10 IR Receiver 19
Figure 3-11 UART Interface 20
Figure 3-12 UART Debug Interface 20
Figure 3-13 RS485 and RS232 Interface 21
Figure 3-14 Video Output Interface 21
Figure 3-15 HDMI Output Interface 22
Figure 3-16 VGA Output Interface 23
Figure 3-17 Audio Line in/out Interface 23
Figure 3-18 Buzzer 24
Figure 3-19 USB3.0 OTG and USB3.0 HOST1 Interface 24
Figure 3-20 USB2.0 HOST2/3 Interface 25
Figure 3-21 RJ45 Interface 26
Figure 3-22 Mini PCIe3.0 Connector 27
Figure 3-23 SATA Interface 27
Figure 3-24 Indicator Lights 28

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Tables

Table 1-1 PCB Functional Interfaces Introduction 3
Table 2-1 Match Relation Between I2C Address and IO Voltage Level 13
Table 3-1 The MIPI_TX Signal Definition 21

1 Overview

1.1 RK3568 Introduction

RK3568 is a quad-core ARM Cortex-A55 low-power and high-performance processor designed for personal mobile Internet devices and AIoT devices.
It provides many powerful embedded hardware engines to optimize the performance of advanced applications. The H.264 video decoder in RK3568 supports 4K@60fps, H.265 video decoder in RK3568 supports 4K@60fps, H.264/H.265 encoder supports 1080p@60fps, and high-quality JPEG codec. RK3568 embedded 3D GPU is fully compatible with OpenGL ES 1.1/2.0/3.2, OpenCL 2.0 and Vulkan 1.1; the special 2D hardware engine maximizes the display performance and can run smoothly at the same time. The built-in NPU supports INT8/INT16 mixed operation. Due to strong compatibility, network models based on a series of frameworks such as TensorFlow/MXNet/PyTorch/Caffe can be easily converted.
There are high-performance external memory interfaces in RK3568 to ensure high-capacity and high-stability system operating memory bandwidth, and it supports multiple memory models such as DDR3, DDR3L, LPDDR3, DDR4, LPDDR4, LPDDR4X, etc.

1.2 RK3568 Block Diagram

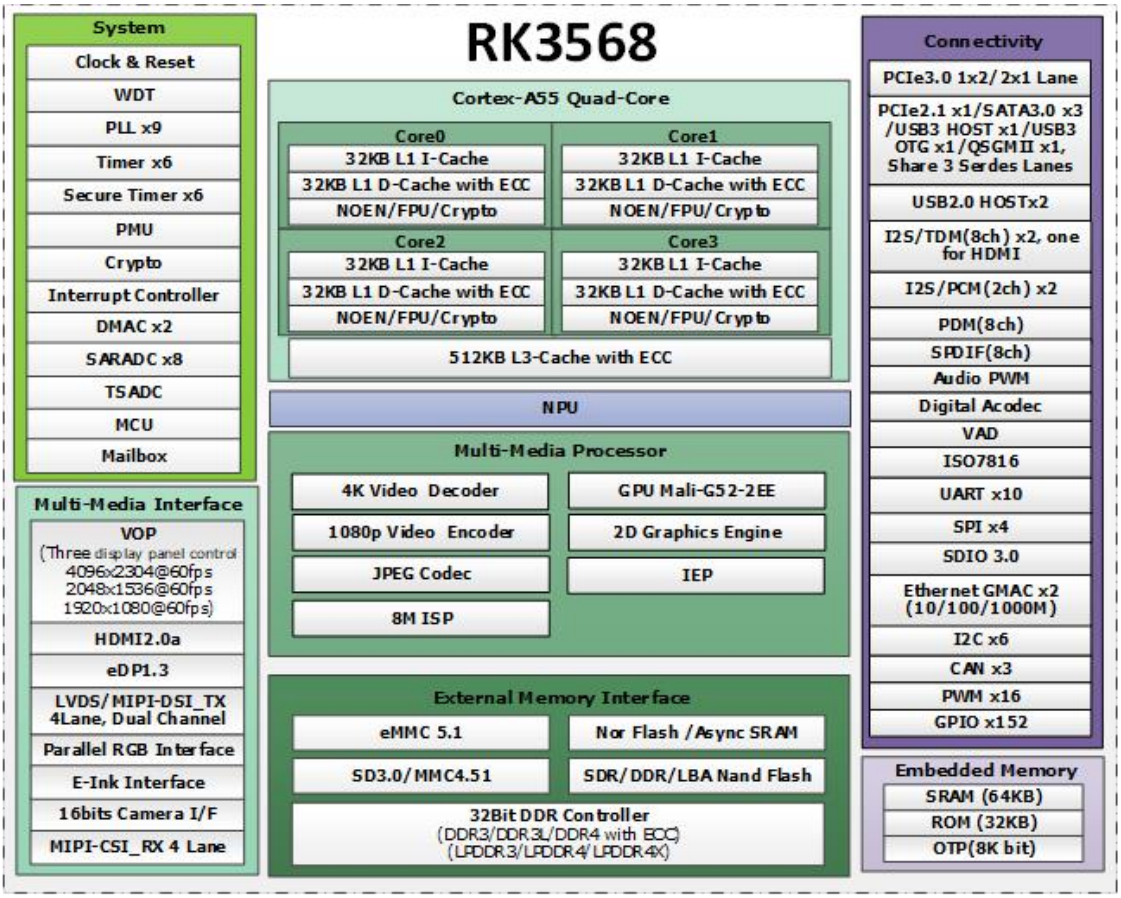

Figure 1-1 RK3568 Block Diagram

1.3 System Introduction

1.3.1 System Diagram

RK3568 NRV DEMO board takes RK3568 as a core chip, uses discrete BUCK and LDO power chips for flexible layout. Uses DDR4, eMMC and SATA/PCIe and other functional external device interfaces for integrating a stable mass production solution. The detailed system block diagram is as follows:

RK3568 NVR DEMO Block Diagram

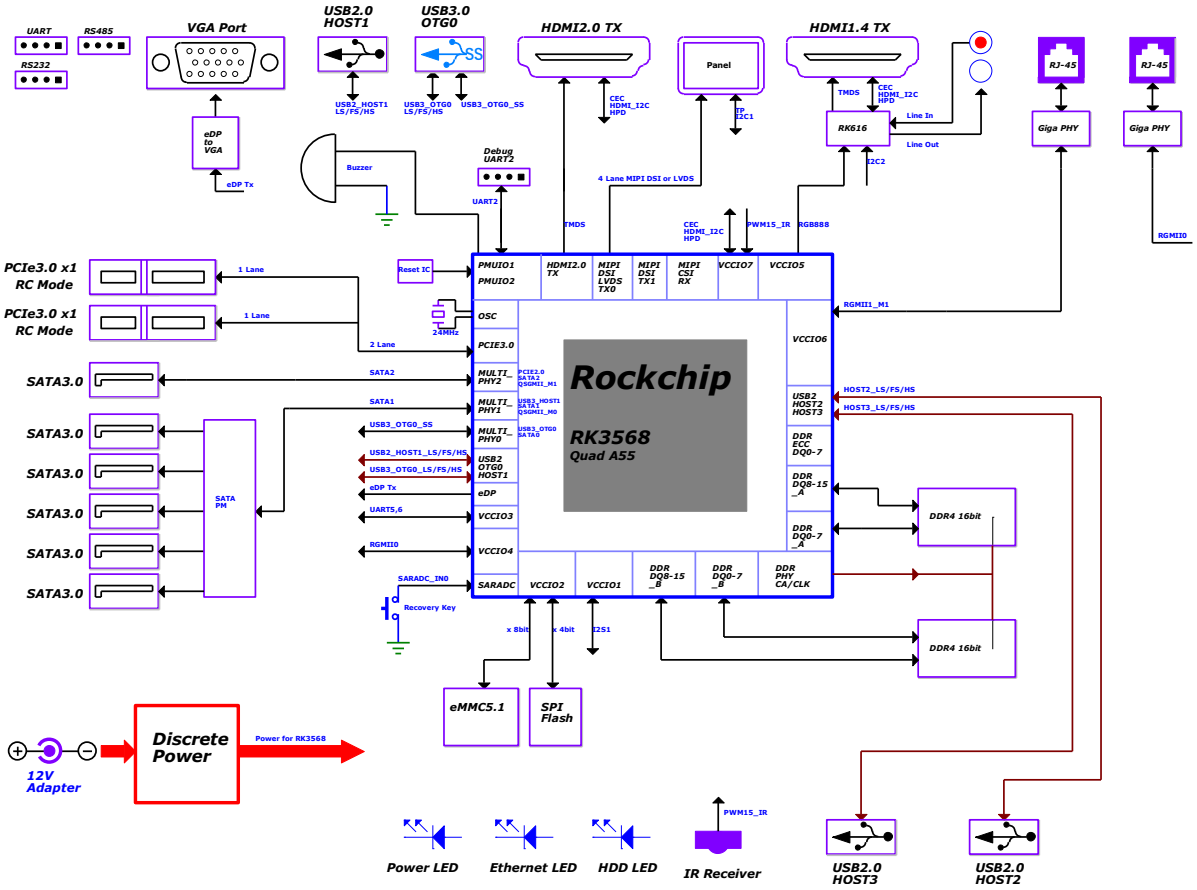

Figure 1-2 RK3568 NVR DEMO System Diagram

1.3.2 Functions Overview

RK3568 NVR DEMO board includes the following functions:

- ✧ DC Power: ① DC 12V adapter power supply interface; ② One ATX main power interface; ③ one ATX + 12V DC interface;
- ✧ USB3.0 OTG: One USB3.0 interface which is used for system firmware upgrading;
- ✧ USB2.0 HOST1: One USB2.0 interface, which can be connected to mouse, U disk, USB HUB and other devices;
- ✧ USB 2.0 HOST2, HOST3: The standard 2.54mm 9pin interface which can be connected as 2 USB interfaces
- ✧ HDMI2.0 OUT: Support up to 4K@60Hz output;
- ✧ HDMI1.4 out: Convert RGB to HDMI output, supporting up to 1080p

- ✧ MIPI DSI/LVDS: Support 4lane MIPI or LVDS signal output, connected via FPC line;
- ✧ VGA OUT: Convert eDP signal to VGA output;
- ✧ Ethernet: Support 2 RGMII 10/100/1000M Ethernet;
- ✧ Audio Interface: Support audio output by a buzzer, line in and line out;
- ✧ SATA Interface: Support six 7pin SATA interfaces, one 4pin SATA power interface;
- ✧ Mini PCIe Interface: Two Mini PCIe, supports users to extend and debug PCIe devices;
- ✧ IR Receive: IR remote control input;
- ✧ LED Indicator Light: Working indicator, Ethernet connection indicator and hard disk connection indicator;
- ✧ System Key: Including Reset, Recover key and reserved MASKROM test point;
- ✧ UART debug: Debug and check log information;
- ✧ UART: Support external UART functional devices;
- ✧ RS232: Support external RS232 bus device;
- ✧ RS485: Support external RS485 bus device;

1.3.3 Functional Interfaces

Table 1-1 PCB Functional Interfaces Introduction

Functions	Useable or not
DDR4(512x16bit, total capacity is 2GB)	YES
eMMC (total capacity is 32GB)	YES
SPI Flash	It is NC by default SPI Flash or eMMC, just choose one
DC 12V Input (default power supply mode)	YES
ATX Power	YES
ATX Main POWER	Jumper resistances are needed, please refer to Section 3.1
USB3.0 OTG	YES
USB2.0 Host(3 Port)	YES
HDMI2.0 OUT	YES
RGB转HDMI1.4 OUT	YES
MIPI DSI/LVDS(4lane)	YES
VGA OUT	YES
RGMII 10M/100M/1000M (2 Port)	YES
Audio (beep、line in/out)	YES
SATA3.0 (6 Port)	YES
Mini PCIe Interface (2 Port)	YES
IR Receive	YES
UART Debug	YES
UART	YES
Indicator Light	YES
System Key	YES

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Functions	Useable or not
RS232	YES
RS485	YES

1.3.4 Functional Modules Layout

The NVR DEMO board functional interfaces layout are showed as follows：

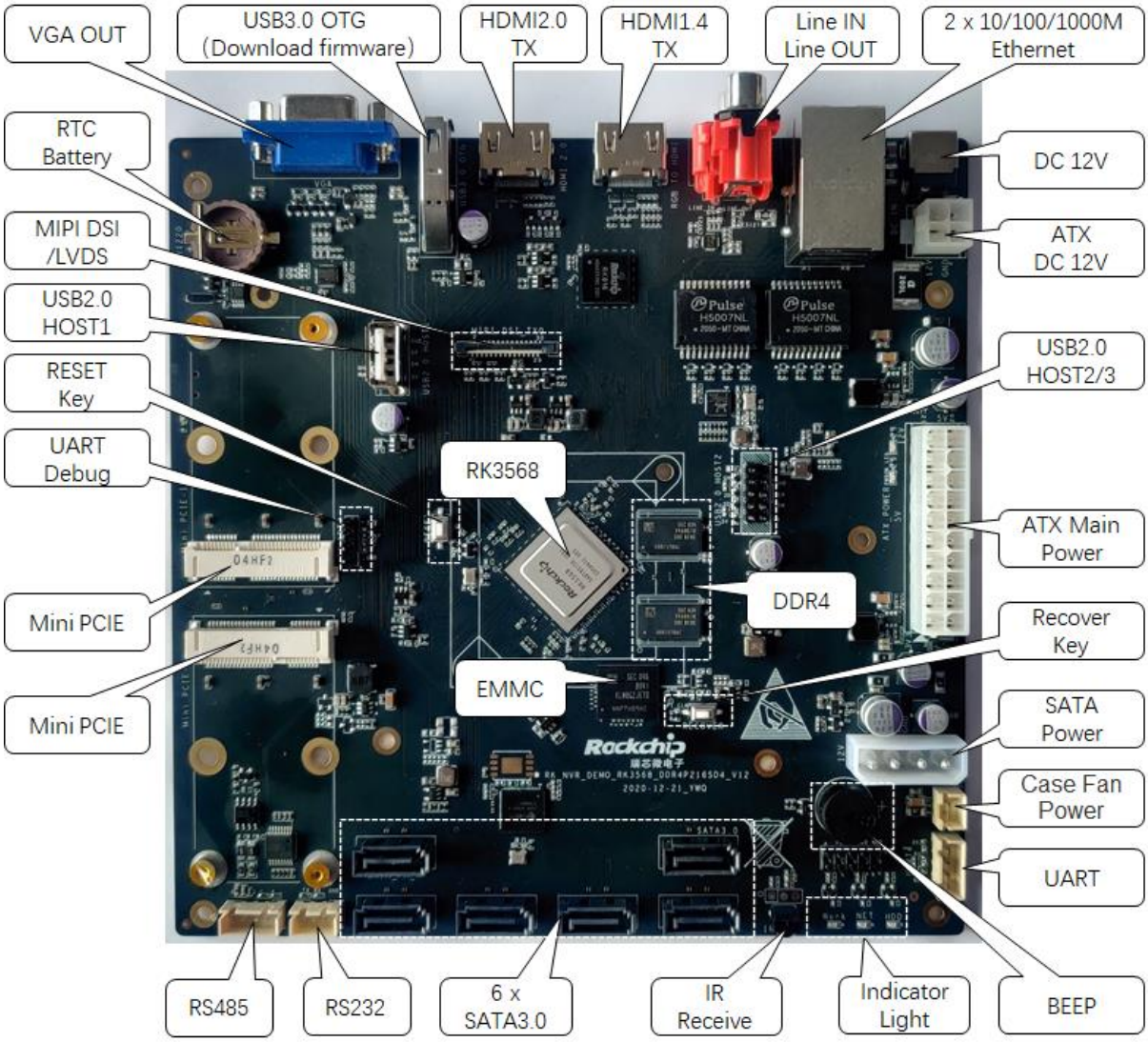

Figure 1-3 NVR DEMO Board Functional Interfaces Layout (TOP Surface)

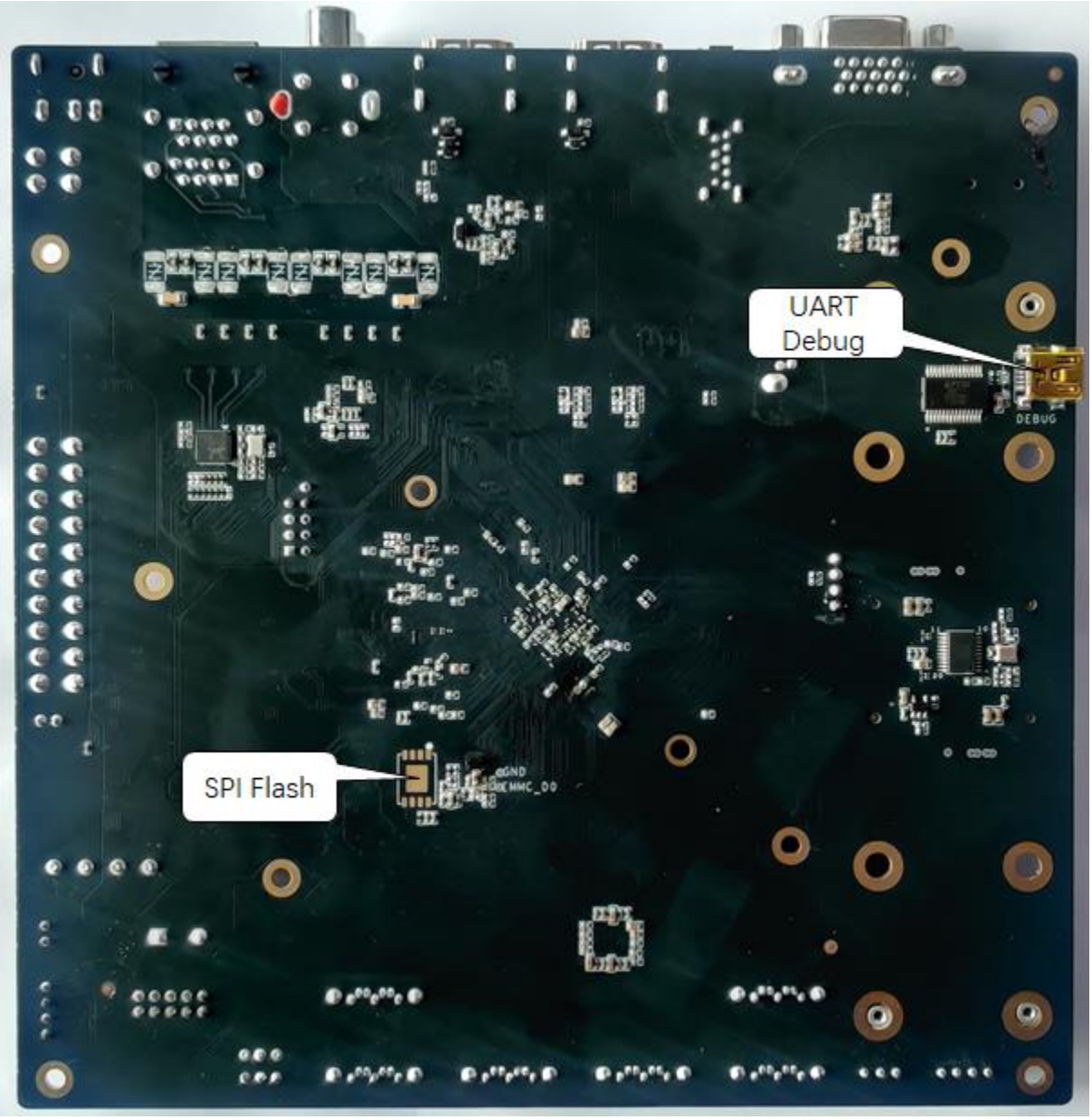

Figure 1-4 NVR DEMO Board Functional Interfaces Layout (Bottom Surface)

1.4 Components

RK3568 NVR DEMO board package includes the following components:

- One RK3568 NVR DEMO board
- One DC adapter: input 100V AC~240V AC, 50Hz, output DC12V DC, 3A

1.5 Power on and off

The way to power on and off the NRV DEMO board will be introduced as follows:

- Power on by the following three ways of power supply

  1. Powered by DC 12V power adapter (Default power supply mode).

2. Powered by ATX DC 12V power
3. Powered by ATX main power, but jumper resistances are required, please refer to Section 3.1 for details

- The way to power off:
  Turn off the DC 12V adapter and the ATX Power supply to power off; if there is a hard disk, you need to power off through software, and then turn off the external power supply, otherwise the hard disk may be damaged.

1.6 Driver Upgrade

1.6.1 USB Driver Installation

It has to install the driver before upgrading the NVR DEMO board driver, the tool is located in: SDK\RKTools\windows\Release_DriverAssitant, open "DriverInstall.exe", click "Driver Installation", to install the driver. If a previous version driver has been installed, please click "Driver Uninstall" and reinstall the driver.
The driver file almost covers all current operating systems which can be supported.

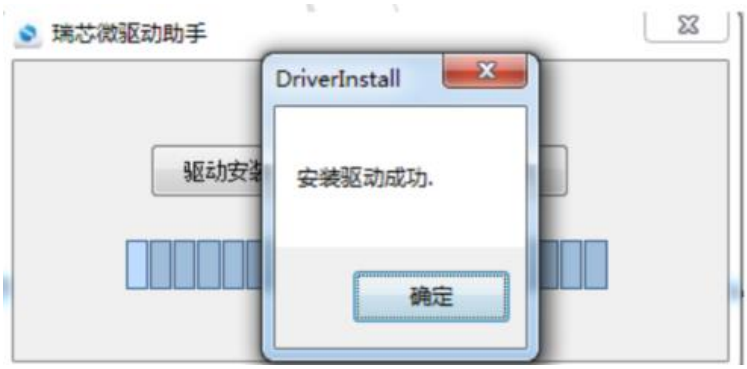

Figure 1-5 Driver Installed Successfully

1.6.2 Driver Upgrade

There are two ways to upgrade the driver of RK3568 NVR DEMO board:
Enter Loader upgrade mode:
Ensure that SARADC_VIN0 is low before the system is powered on, press and hold RECOVER key of the board, the system will enter the Loader state.
Detailed steps are as follows:
1. Connect the USB3.0 OTG port to a PC, press and hold RECOVER key of the DEMO board.
2. The DEMO board is powered by 12V. If it has been powered on, press the reset key.
3. After the flashing tool shows that a Loader device is found, release the RECOVER key.
4. Select Loader, Parameter, Uboot, Recovery and other files on the flashing tool.
5. Click Execute to enter the upgrade state. The right side of the tool is the progress display bar, which displays the download progress and verification status.

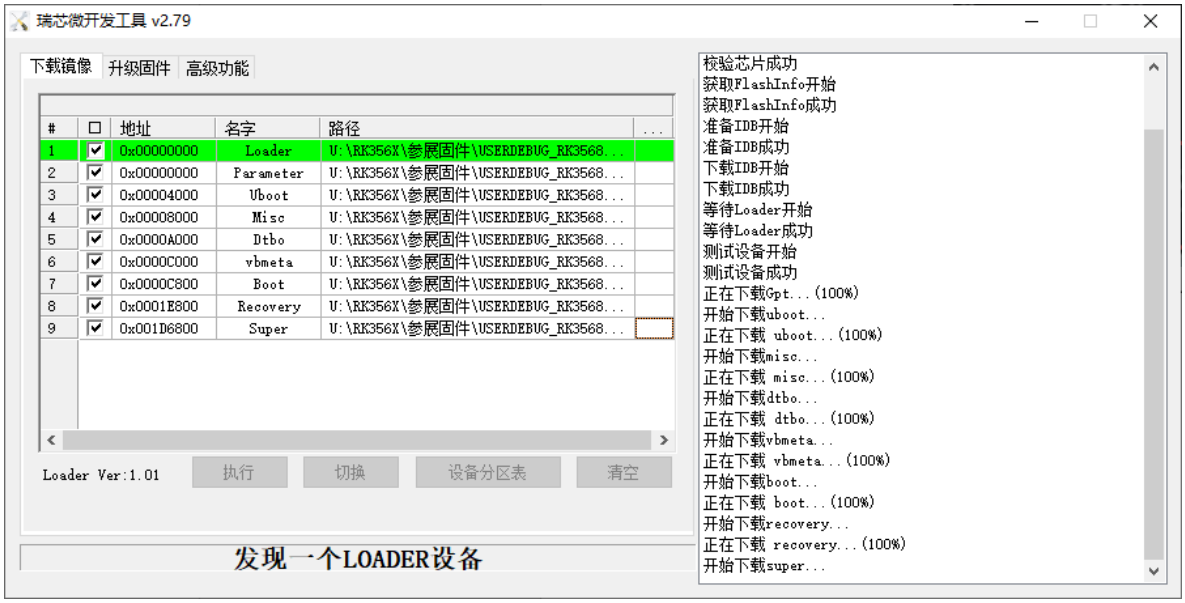

Figure 1-6 Loader Flashing Mode

If you fail to upgrade in Loader mode, please enter Maskrom upgrade mode:

Before the system is powered on, connect eMMC_D0 to ground, causing eMMC to fail to boot and enter the Maskrom state.

Detailed steps are as follows:

1. Connect the USB3.0 OTG port to a PC, connect eMMC_D0 to GND by a tweezers
2. The DEMO board is powered by 12V. If it has been powered on, press the reset button.
3. After the flashing tool shows that a Maskrom device is found, release the connection of eMMC_D0 to GND. Note that in the Maskrom state, you need to select the corresponding Loader options to finish the upgrading.
4. Select Loader, Parameter, Uboot, Recovery and other files on the flashing tool.
5. Click Execute to enter the upgrade state. The right side of the tool is the progress display bar, which displays the download progress and verification status.

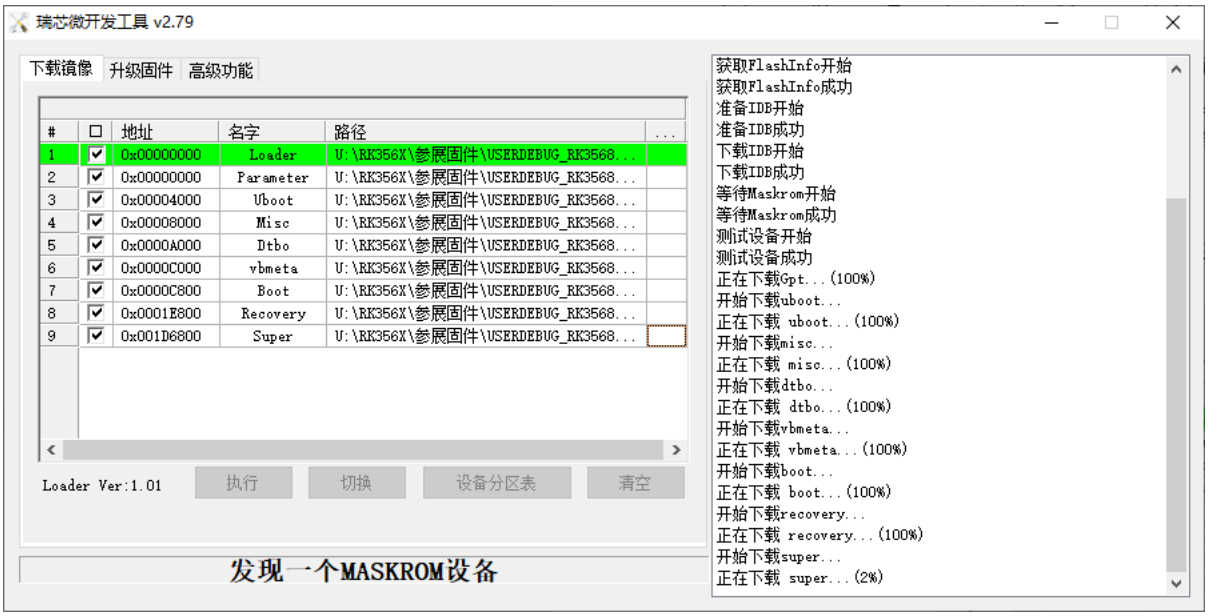

Figure 1-7 Maskrom Flashing Mode

1.7 Serial Port Debugging

1.7.1 Serial Port Debugging Tool

Connect the UART Debug interface of the DEMO board to a PC through a serial port adapter board, to get the current COM number in the PC device manager.

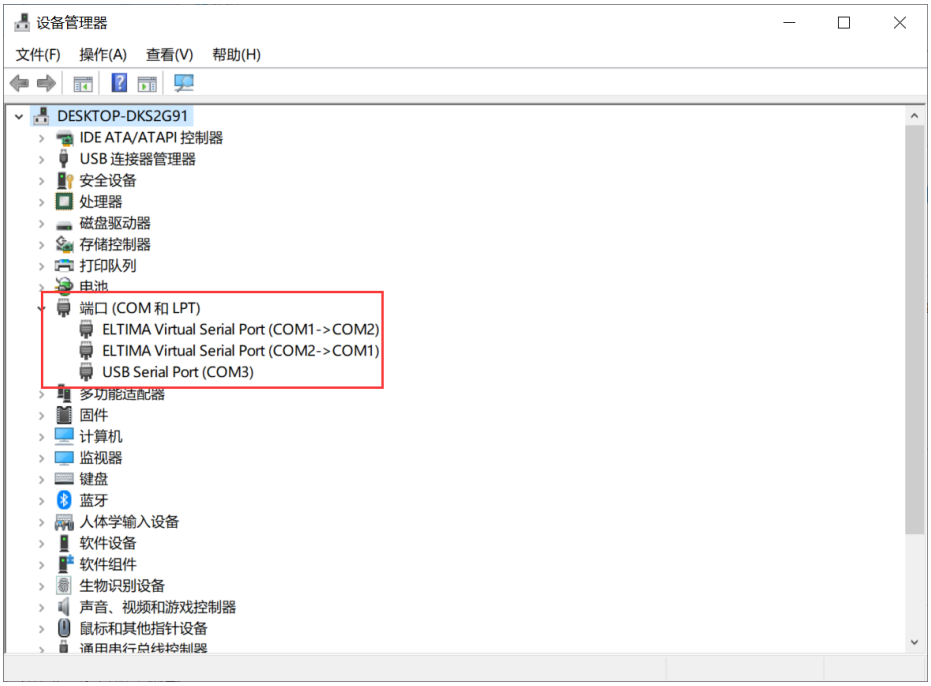

Figure 1-8 Current COM Number

Open the serial port debug tool. Firstly, enter Session interface; Secondly select “Serial” under Connection type; thirdly, select the corresponding serial port number and change the baud rate to 1.5M (RK3568 supports 1.5M baud rate by default), and finally click the “open” button to enter the serial debugging

interface.

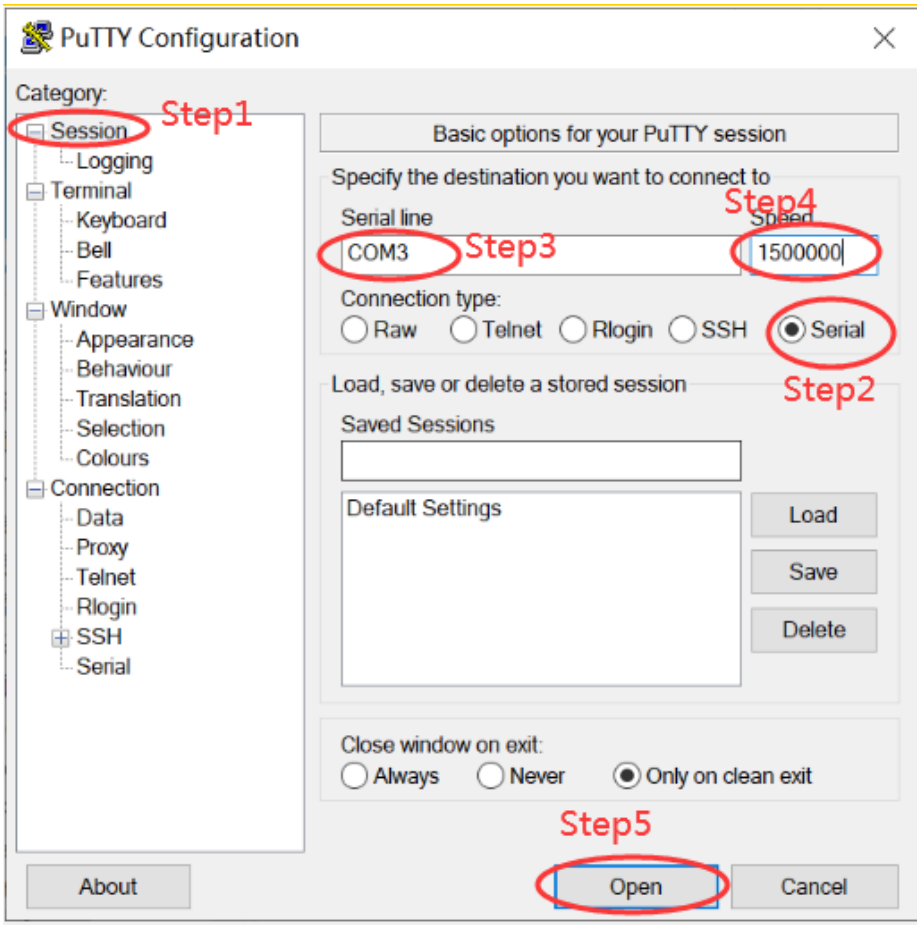

Figure 1-9 Serial Port Tool Configuration Interface

Figure 1-10 Serial Port Tool Debugging Interface

1.7.2 ADB Debug

1. Ensure that the driver is installed successfully, and the PC is connected to the USB3.0 OTG port of the DEMO board;

2. Power on the DEMO board and boot into the system;
3. On the PC, Click start ---run ---cmd, enter the directory where the adb.exe tool is located, and input “adb devices” if the connected device is searched, indicating that the connection is successful;
4. Input "adb shell" to enter ADB debug.

Figure 1-11 ADB Connected Successfully

2 Hardware Introduction

2.1 The Picture of NVR DEMO Board

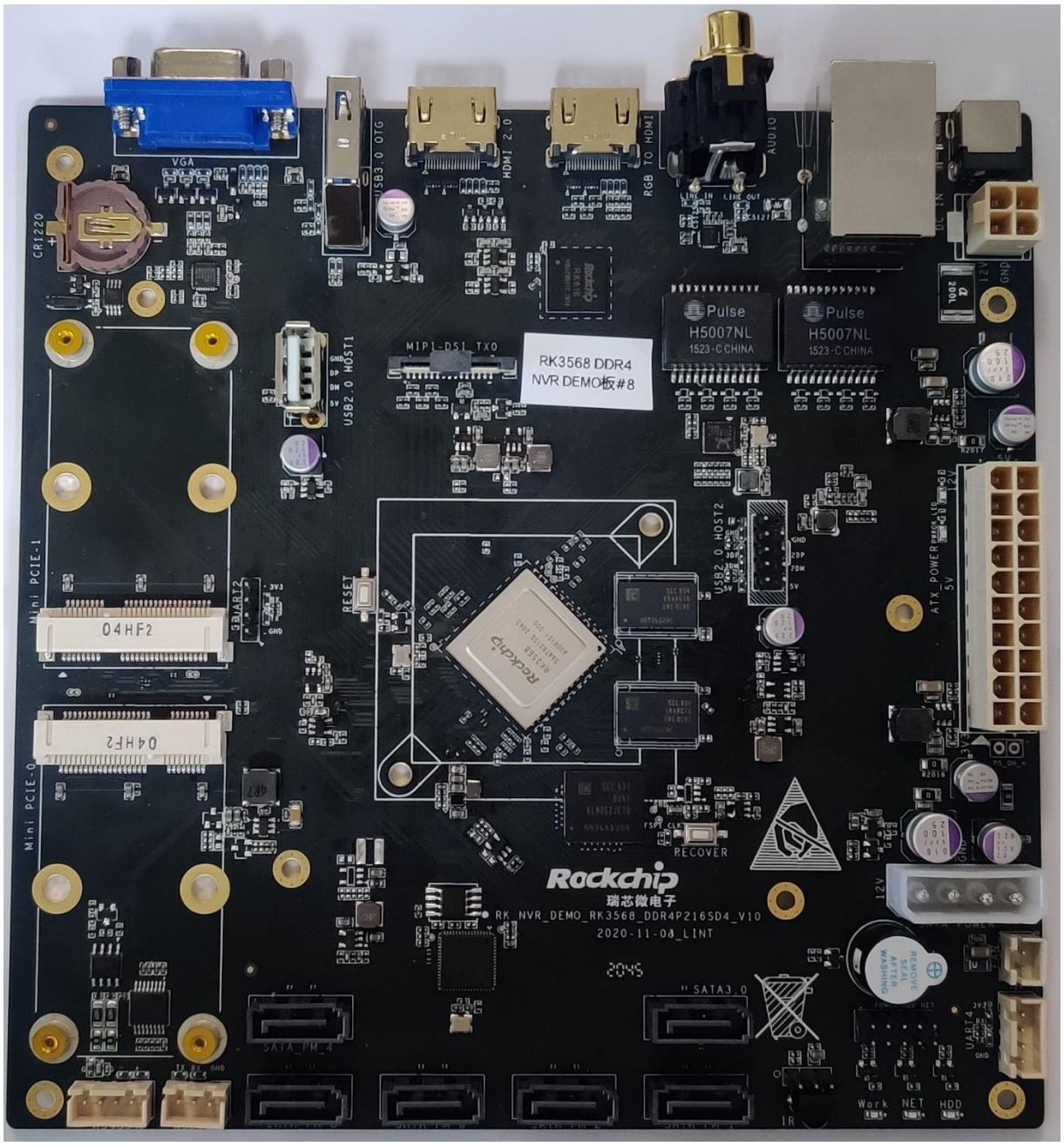

Figure 2-1 The Picture of NVR DEMO Board

2.2 Power Diagram

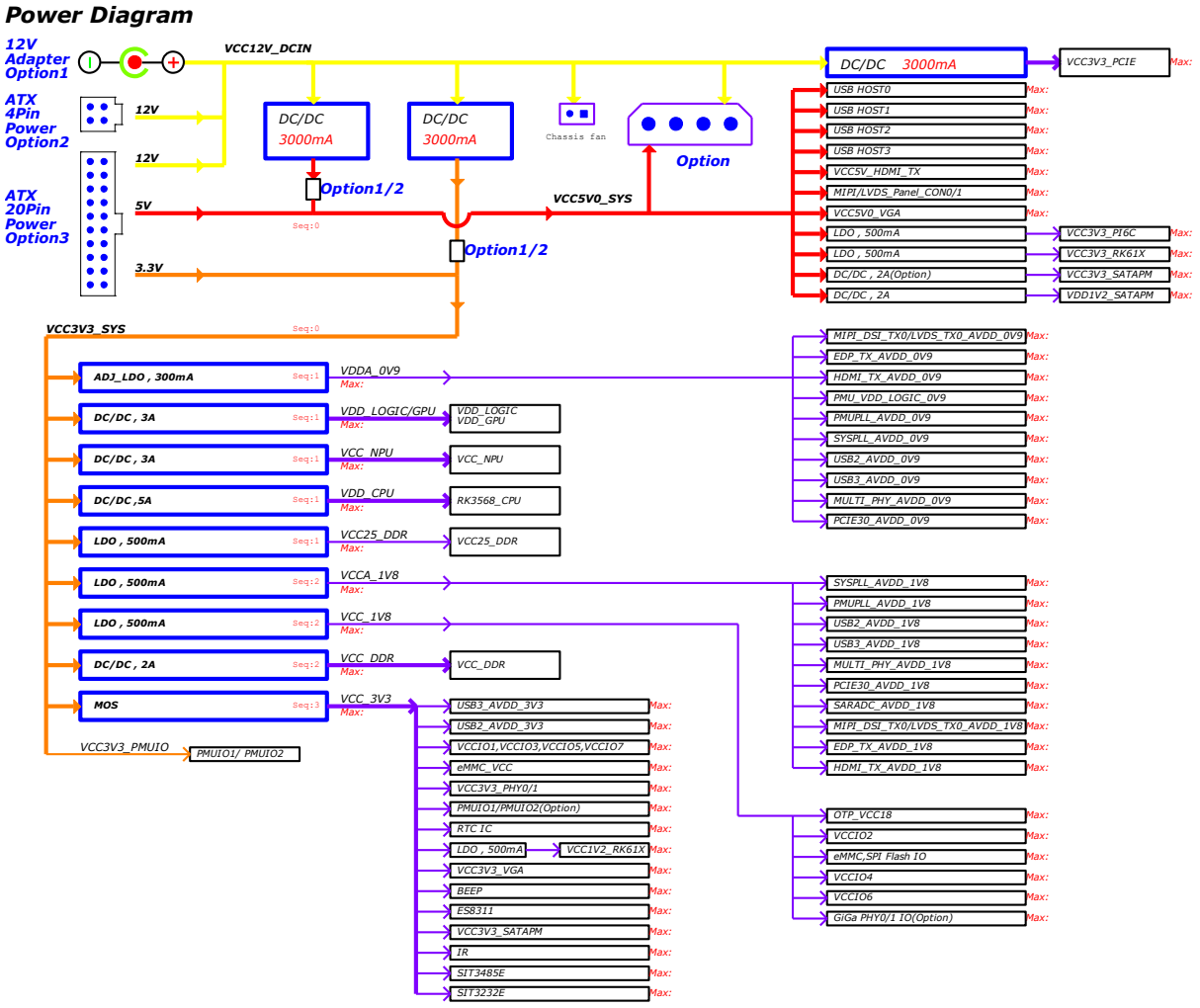

Figure 2-2 RK3568 NVR DEMO Board Power Diagram

2.3 I2C Address

The DEMO board reserves a reach peripheral interfaces. I2C peripherals debugging will relate to I2C channel multiplexing. Table 2-1 shows the match relation between I2C address of the DEMO board and voltage level, to avoid address conflicts and mismatched.

Table 2-1 Match Relation Between I2C Address and IO Voltage Level

I2C Channel	Devices	I2C address	Power domain
I2C0	TCS4525（DCDC）	0X1C	3.3V
I2C1	HYM8563TS（RTC）	Read：0XA3 Write：0XA2	3.3V
I2C1	MIPI CSI/LVDS	TBD	3.3V
I2C1	RTD2166(VGA)		3.3V
I2C3	RK616(HDMI)	0X50	3.3V
I2C3	ES8311(Audio)	CE=0,0X18 CE=1,0X19	3.3V

Note: when using an extension board, please ensure that the I²C address of the extension board does not conflict with the I²C address of the DEMO board

2.4 Extension Connectors

Users may use extension boards in practical use. There are several types of connectors for DEMO board: J5200 is a vertical double-row 30pins with 0.5mm pins and 1mm pitch (corresponding to the FPC cable with 0.5mm pitch, 30pin), the dimensions are as follows:

P数	A	B	C	D	P数	A	B	C	D
4	1.500	2.570	8.400	4.650	35	17.000	18.070	23.900	5.150
5	2.000	3.070	8.900	4.650	36	17.500	18.570	24.400	5.150
6	2.500	3.570	9.400	4.650	37	18.000	19.070	24.900	5.150
7	3.000	4.070	9.900	4.650	38	18.500	19.570	25.400	5.150
8	3.500	4.570	10.400	4.650	39	19.000	20.070	25.900	5.150
9	4.000	5.070	10.900	4.650	40	19.500	20.570	26.400	5.150
10	4.500	5.570	11.400	4.650	41	20.000	21.070	26.900	5.150
11	5.000	6.070	11.900	4.650	42	20.500	21.570	27.400	5.150
12	5.500	6.570	12.400	4.650	43	21.000	22.070	27.900	5.150
13	6.000	7.070	12.900	4.650	44	21.500	22.570	28.400	5.150
14	6.500	7.570	13.400	4.650	45	22.000	23.070	28.900	5.150
15	7.000	8.070	13.900	4.650	46	22.500	23.570	29.400	5.150
16	7.500	8.570	14.400	4.650	47	23.000	24.070	29.900	5.150
17	8.000	9.070	14.900	4.650	48	23.500	24.570	30.400	5.150
18	8.500	9.570	15.400	4.650	49	24.000	25.070	30.900	5.150
19	9.000	10.070	15.900	4.650	50	24.500	25.570	31.400	5.150
20	9.500	10.570	16.400	4.650	51	25.000	26.070	31.900	5.150
21	10.000	11.070	16.900	4.650	52	25.500	26.570	32.400	5.150
22	10.500	11.570	17.400	4.650	53	26.000	27.070	32.900	5.150
23	11.000	12.070	17.900	4.650	54	26.500	27.570	33.400	5.150
24	11.500	12.570	18.400	4.650	55	27.000	28.070	33.900	5.150
25	12.000	13.070	18.900	4.650	56	27.500	28.570	34.400	5.150
26	12.500	13.570	19.400	4.650	57	28.000	29.070	34.900	5.150
27	13.000	14.070	19.900	4.650	58	28.500	29.570	35.400	5.150
28	13.500	14.570	20.400	4.650	59	29.000	30.070	35.900	5.150
29	14.000	15.070	20.900	4.650	60	29.500	30.570	36.400	5.150
30	14.500	15.570	21.400	5.150	61	30.000	31.070	36.900	5.150
31	15.000	16.070	21.900	5.150	62	30.500	31.570	37.400	5.150
32	15.500	16.570	22.400	5.150	63	31.000	32.070	37.900	5.150
33	16.000	17.070	22.900	5.150	64	31.500	32.570	38.400	5.150
34	16.500	17.570	23.400	5.150					

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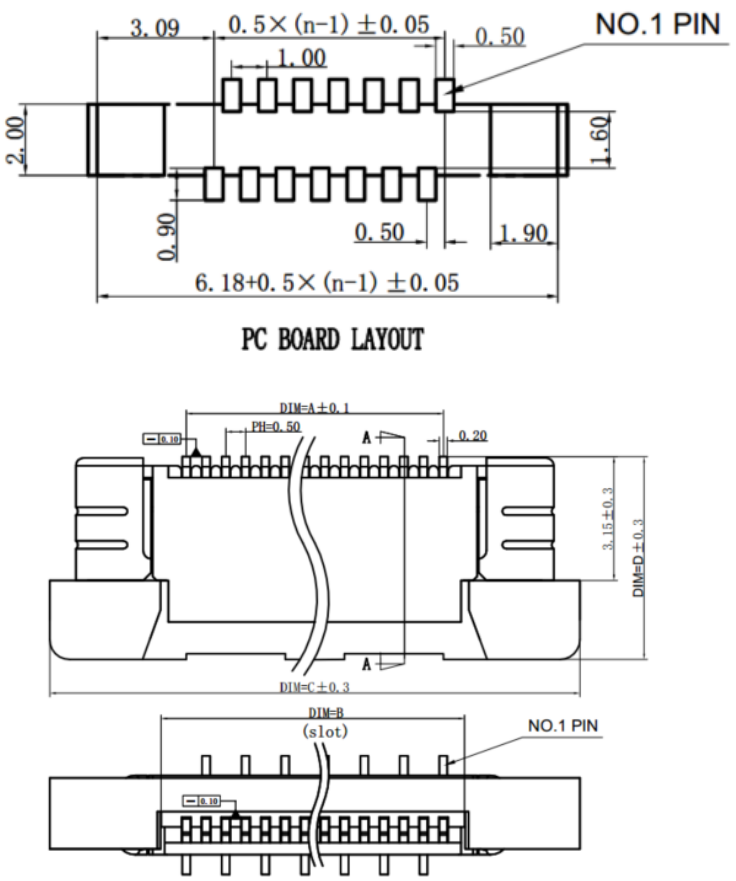

Figure 2-3 Vertical Double Row 30 PIN PCB Package with 1mm Pitch

2.5 Reference Materials

The reference diagram and PCB version information of the NVR DEMO board are as follows
RK_NVR_DEMO_RK3568_DDR4P216SD4_V12_20201217.DSN
RK_NVR_DEMO_RK3568_DDR4P216SD4_V12_20201218.brd

3 Modules Introduction

3.1 Power Input

The following are three ways of power input:
1. Only 1 external hard disk: powered by an adapter to input 12V/3A power; if you connect multiple hard disks, the adapter can also be used for power supply when the total power does not exceed the maximum power of the adapter.
2. 6 external hard disks: powered by ATX +12V DC Power (not provided by default)
3. 6 external hard disks: powered by ATX Main Power (not provided by default)
The power adapter and ATX +12V DC Power input 12V power, after going through the front-end step-down converter (buck) power, will get the system power VCC5V0_SYS and VCC3V3_SYS which will provide to multiple discrete DCDC and LDOs to output different voltages for system.
You have to disconnect the resistor R2016/R2017 when using ATX Main Power, and directly provide VCC12V_DCIN, system power VCC5V0_SYS and VCC3V3_SYS, and then provide the system voltage to multiple discrete DCDCs and LDOs to output different voltages for system.
Power adapter input port and front-end Buck converter:

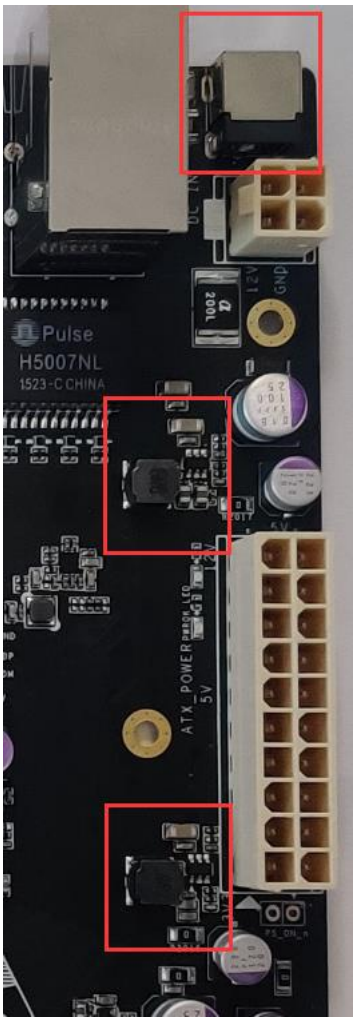

Figure 3-1 DC12V Input and Front-end Buck Converter

One 20pin ATX Main POWER interface, which complies with Version1.0 standard; one 4pin ATX +12V DC interface.

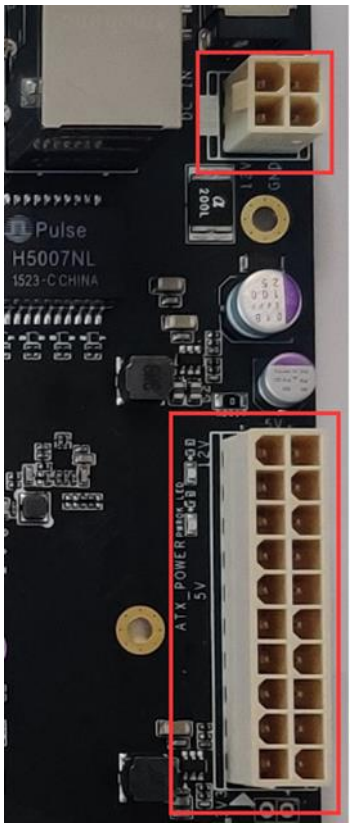

Figure 3-2 ATX Main POWER and ATX +12V DC Interface

The 20pin ATX Main POWER interface signals are as follows:

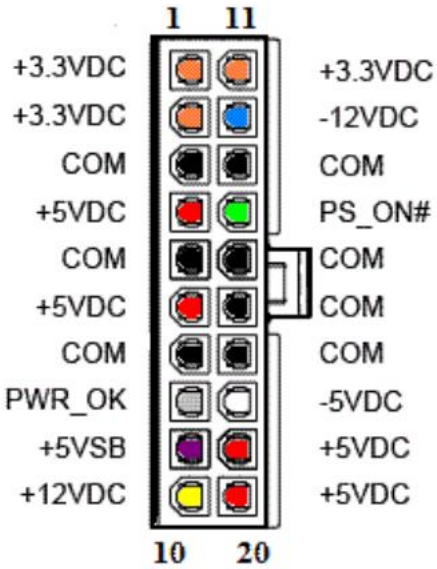

Figure 3-3 20pin ATX Main POWER Version1.0 Interface Signal

The 4pin ATX +12V DC interface signals are as follows:

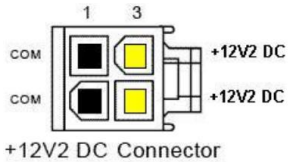

Figure 3-4 4pin ATX +12V DC Interface Signal

3.2 Memory

1.eMMC

- The memory on the DEMO board is eMMC FLASH with the capacity of 32GB by default
- There is a Maskrom test point reserved on the DEMO board for entering Maskrom to upgrade firmware

2.SPI Flash

- The DEMO board reserves SPI Flash location and there is a Maskrom test point reserved on the DEMO board for entering Maskrom to upgrade firmware

2.DDR

- The DDR on the DEMO board uses two 512M×16bit DDR4, with a total capacity of 2GB

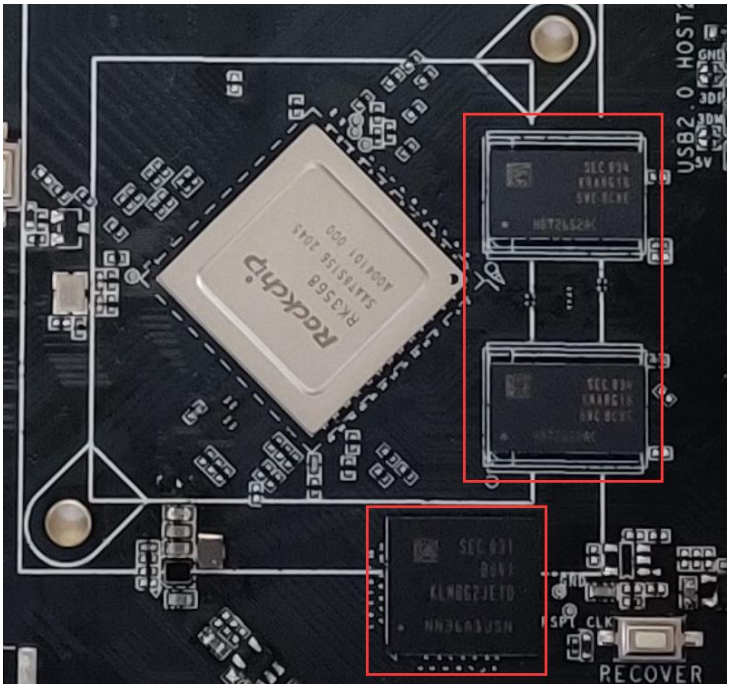

Figure 3-5 DDR4 and eMMC

The test point location of NVR DEMO board for entering Maskrom flashing mode:

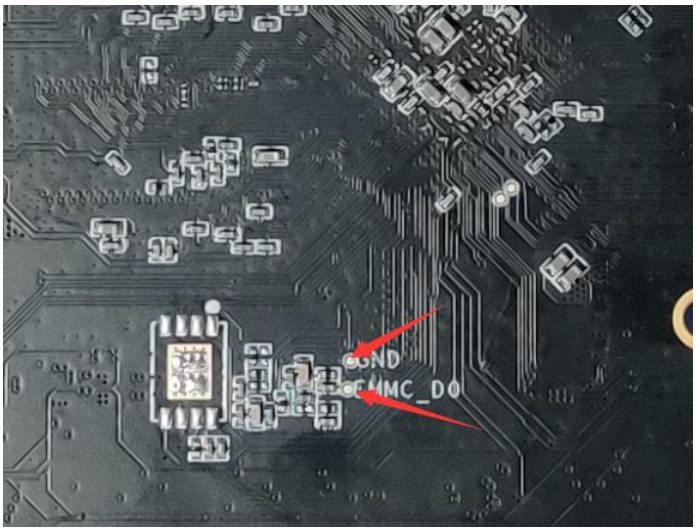

Figure 3-6 The Location theTest Point When Using eMMC Entering Maskrom Flashing Mode

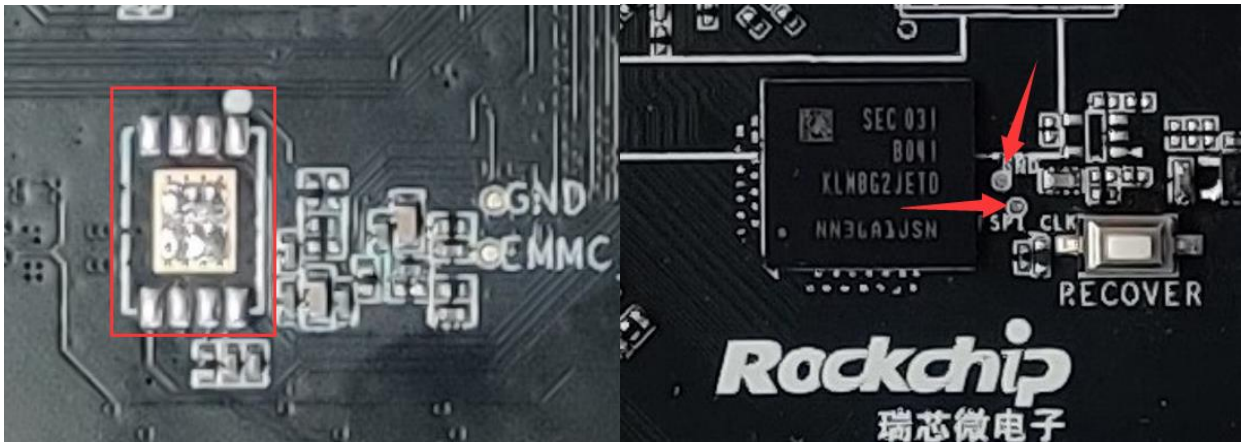

Figure 3-7 The Location of SPI Nand and Test Point When Using SPI Nand Entering Maskrom

3.3 RTC Circuit

The RTC circuit is based on HYM8563TS chip, which can be powered by the DEMO Board or built-in button battery. It can keep providing accurate time even when the board is powered off, and communicate with the controller through I2C signal.

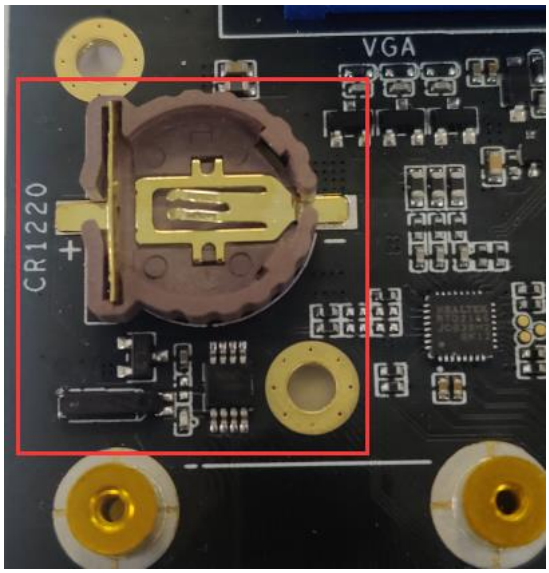

Figure 3-8 RTC Circuit

3.4 Keys Input

The DEMO board takes SARADC_VIN0 as RECOVER detection port, supports 10-bit resolution, and it will go to loader flashing mode by the RECOVER key; in addition, there is also a RESET key on the board, which is convenient for resetting and restarting the device through hardware; Keys location are as follows:

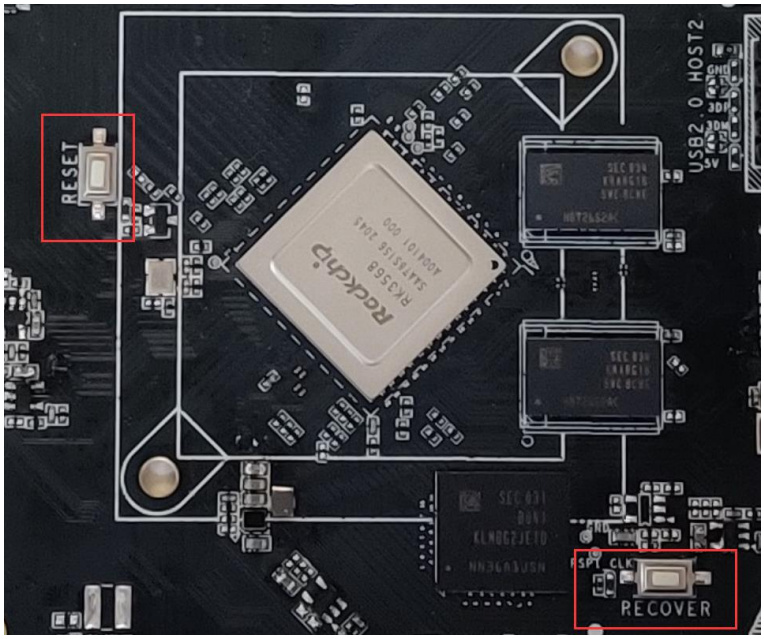

Figure 3-9 Keys Location

3.5 Infrared Receiver

The small infrared receiver on the DEMO board is general model IRM3638, with a center frequency of 38KHz.

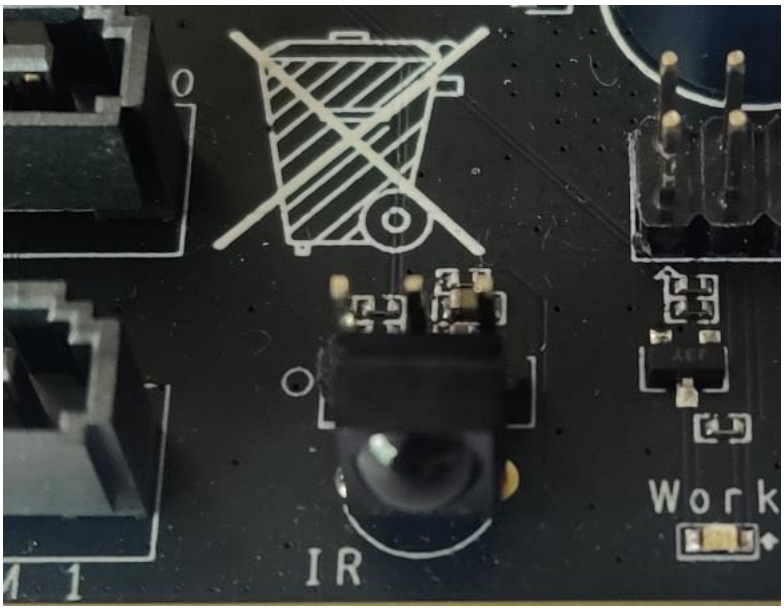

Figure 3-10 IR Receiver

3.6 UART Interface

The DEMO board reserves UART interfaces, which communicate with the controller through UART4. The external standard 4 PINS 2.54mm male socket is convenient for UART peripheral debugging.

Figure 3-11 UART Interface

3.7 UART Debug Interface

The DEMO board provides a serial port for debugging, it is UART2 by default, use FT232RL chip to covert UART signal to USB signal, providing a mini USB interface to connect with PC and the baud rate is 1.5M by default.

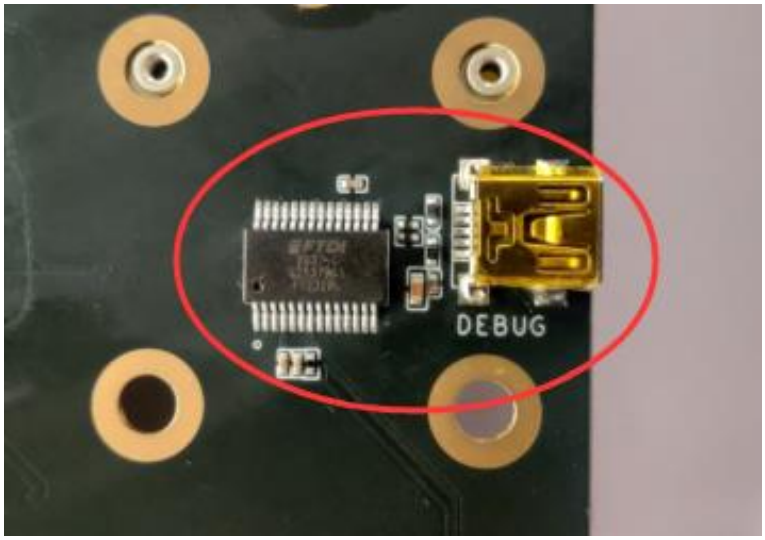

Figure 3-12 UART Debug Interface

3.8 RS485 and RS232 Interface

The RS485 bus on the DEMO board is based on SIT3485E chip, and communicates with the controller through UART6 serial port, providing an external standard 4 PINS 2.54mm male socket for debugging; RS232 bus is based on SIT3232E chip and communicates with the controller through UART5 serial port, provides an external standard 3 PINS 2.54mm male socket for debugging.

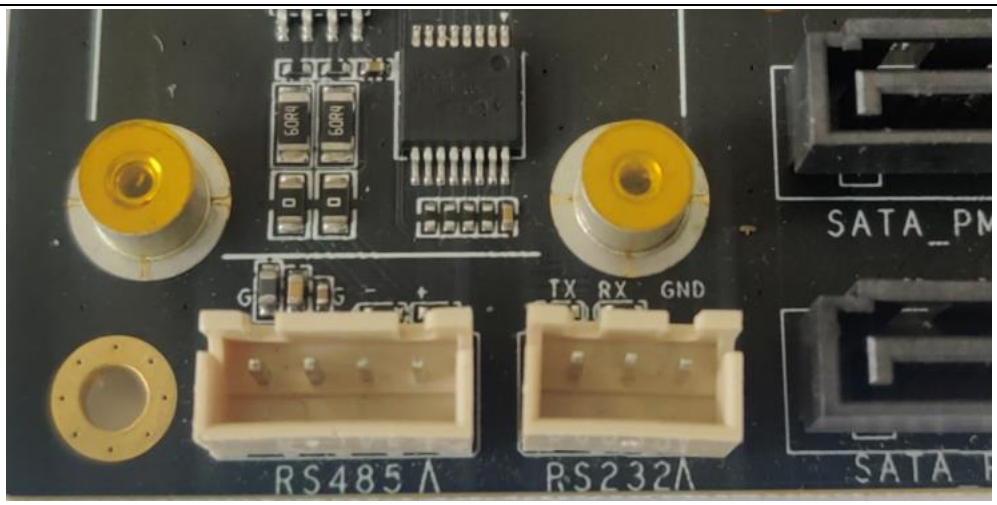

Figure 3-13 RS485 and RS232 Interface

3.9 MIPI/LVDS Output Interface

The MIPI/LVDS output interface uses a vertical connector with a pitch of 1mm.

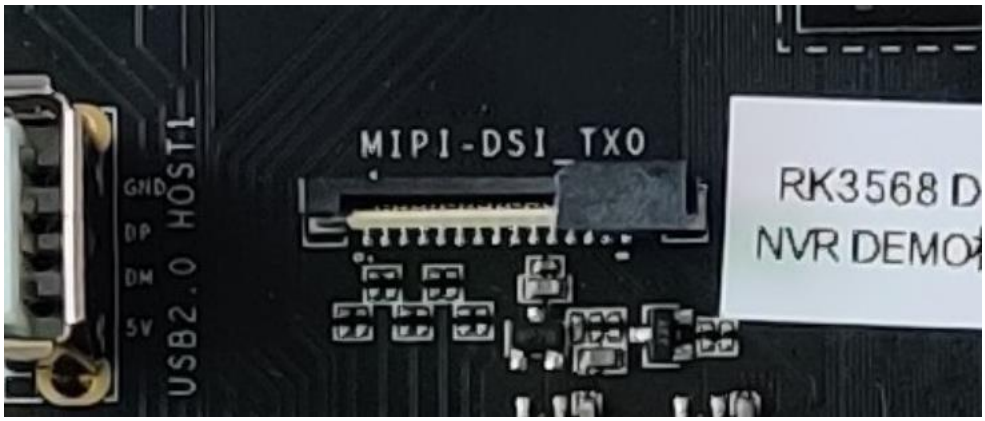

Figure 3-14 Video Output Interface

The MIPI_TX interface signal sequence is as follows:

Table 3-1 The MIPI_TX Signal Definition

1	GND	
2		MIPI_DSI_TX0_D0N
3	MIPI_DSI_TX0_D0P	
4		GND
5	MIPI_DSI_TX0_D1N	
6		MIPI_DSI_TX0_D1P
7	GND	
8		MIPI_DSI_TX0_CLKN
9	MIPI_DSI_TX0_CLKP	
10		GND
11	MIPI_DSI_TX0_D2N	
12		MIPI_DSI_TX0_D2P
13	GND	
14		MIPI_DSI_TX0_D3N

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15	MIPI_DSI_TX0_D3P	
16		GND
17	LCD0_BL_PWM14_M0	
18		NC
19	VCC3V3_LCD0	
20		NC
21	NC	
22		LCD0_PWREN_H
23	I2C_SCL_TP0	
24		ISC_SDA_TP0
25	TP_INT_L	
26		TP_RST_L
27	GND	
28		VCC5V0_SYS
29	VCC5V0_SYS	
30		VCC5V0_SYS

3.10 HDMI Output Interface

The DEMO board supports HDMI output, the left interface outputs HDMI signals by controller directly, supporting HDMI2.0 protocol, the right interface uses RK616 chip to convert RGB to HDMI signal, supporting HDMI1.4 protocol, and the output connectors are HDMI A interfaces.

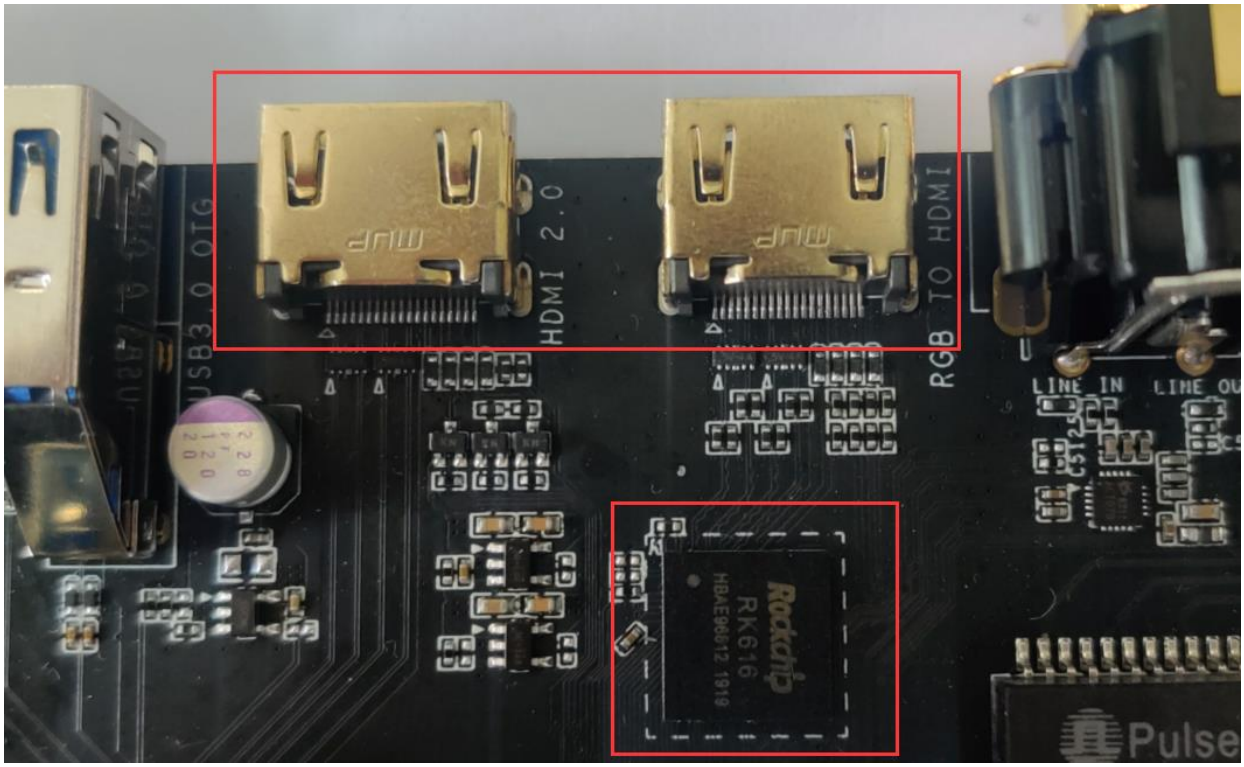

Figure 3-15 HDMI Output Interface

3.11 VGA Output Interface

The DEMO board converts eDP to VGA signal through RTD2166 chip, and the external standard VGA socket is convenient to connect with VGA monitor.

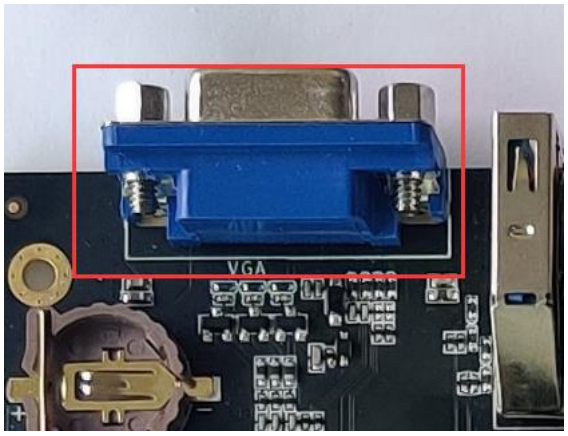

Figure 3-16 VGA Output Interface

3.12 Audio Input and Output Interface

The DEMO board convert RK3568 I2S to Line in/out signal by ES8311 chip by default, or output Line in/out signal by RK616; control the buzzer to output audio by another GPIO port.

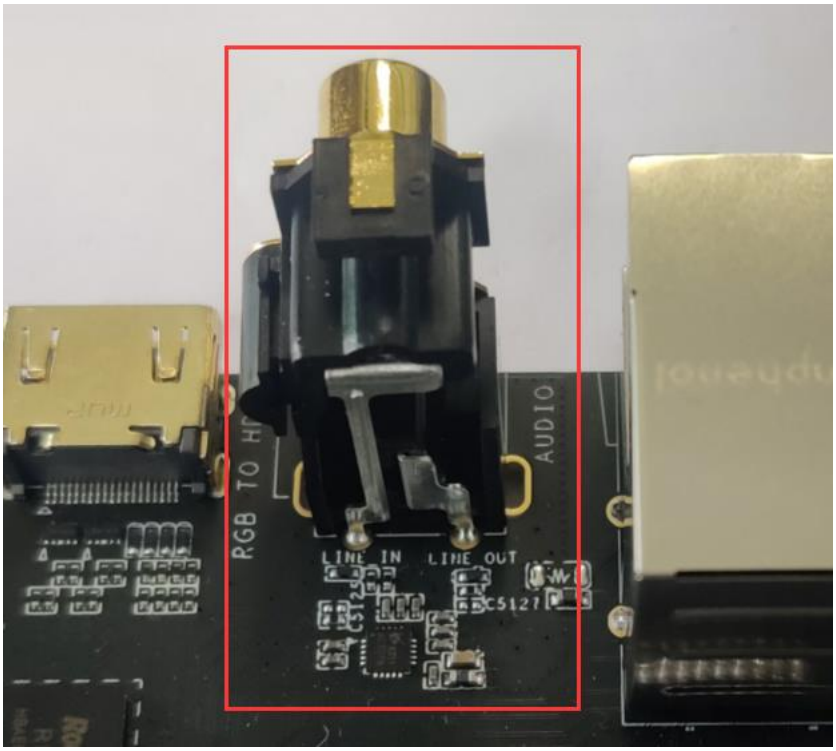

Figure 3-17 Audio Line in/out Interface

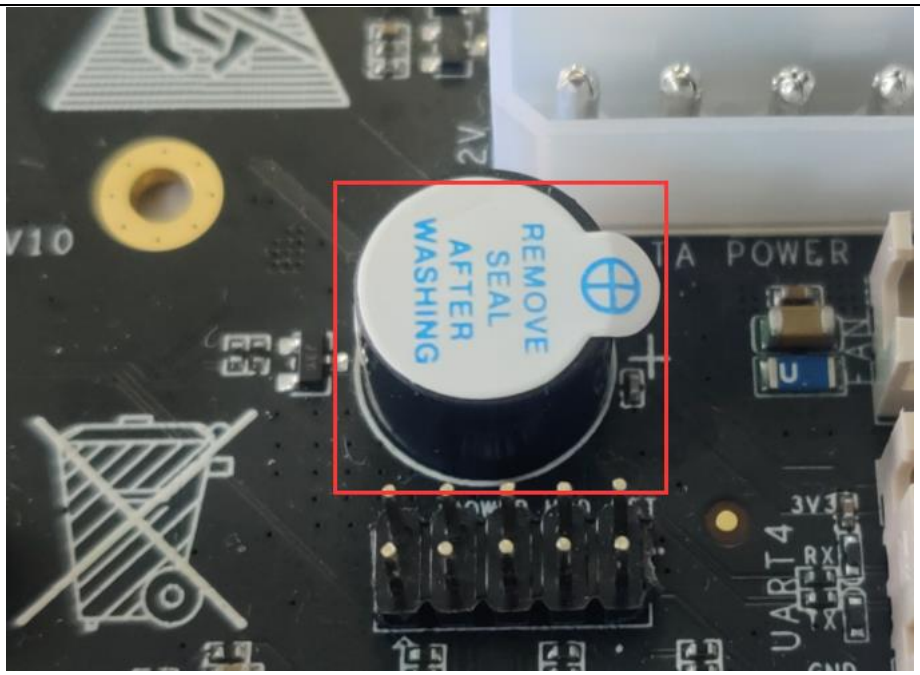

Figure 3-18 Buzzer

3.13 USB OTG/HOST Interface

The DEMO board with USB OTG and USB HOST interface:

USB3.0 OTG consists of 2 USB3.0 high-speed signal lines and one OTG signal line, and is connected to USB3.0 standard-A interface, and is backward compatible with USB 2.0 specifications. Therefore, this interface can be used to download and upgrade firmware, or as a USB3.0 HOST.

USB2.0 HOST1 uses USB2.0 Standard-A interface and standard 4 PIN 2.54mm male dual layout. You can directly use the standard USB2.0 A port to communicate with peripherals, or transfer the 4pin to standard USB2.0 A through an adapter cable.

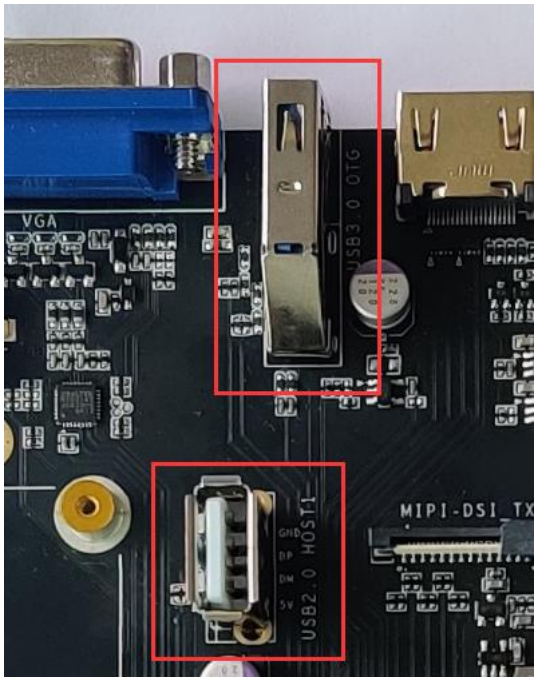

Figure 3-19 USB3.0 OTG and USB3.0 HOST1 Interface

USB2.0 HOST2/3 is a standard 9 PIN 2.54mm male socket, which can be convert to 2 standard USB2.0 A ports by an adapter cable, and convenient for extension.

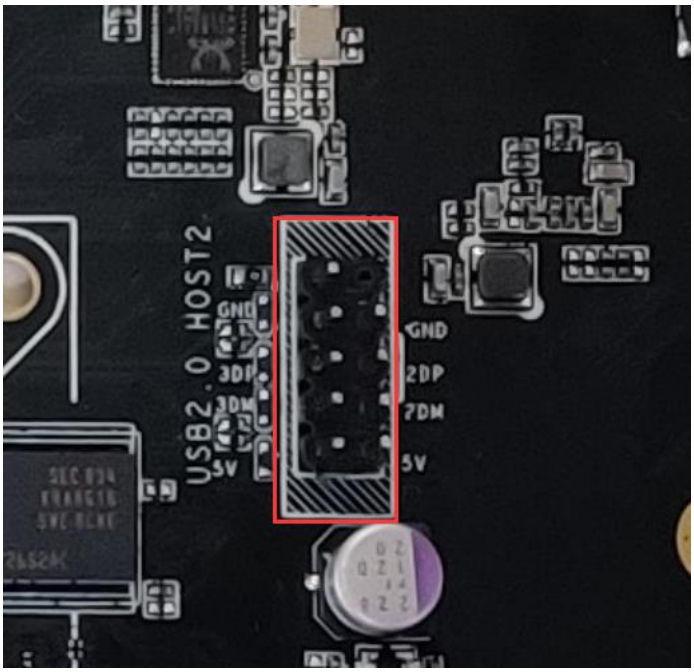

Figure 3-20 USB2.0 HOST2/3 Interface

3.14 Ethernet Interface

The DEMO board supports two RJ45 interfaces and provide dual Gigabit Ethernet connections. Both channels use the Gigabit Ethernet MAC integrated inside RK3568 and are connected to an external PHY chip with the model of RTL8211F-CG, and the features are as follows:

- Compatible with IEEE802.3 standard, support full-duplex and half-duplex operation, support cross detection and self-adaptation.
- Support 10/100/1000M data rate.
- The interface takes dual RJ45 interface units without isolation transformer.

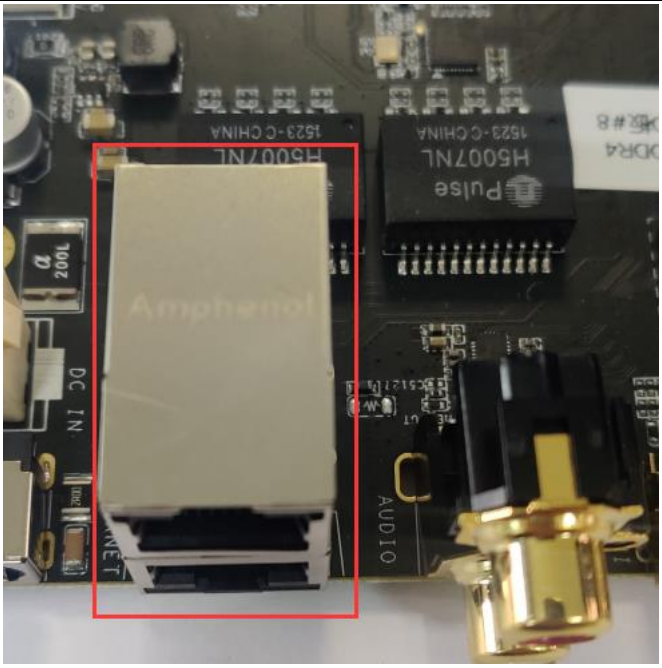

Figure 3-21 RJ45 Interface

3.15 Mini PCIe Interface

The DEMO board uses two standard Mini PCIe3.0 connectors which is compliant with standard PCI Express Mini Card version 1.2 protocol, you can install external PCIe board for communication, for example, external Toybrick TB-RK1808M0 Mini-PCIe computing card to enhance computing capability.

- Working mode: Root Complex (RC).
- Supports 1 lane data interface.
- The 100MHz clock is provided by an external clock chip.

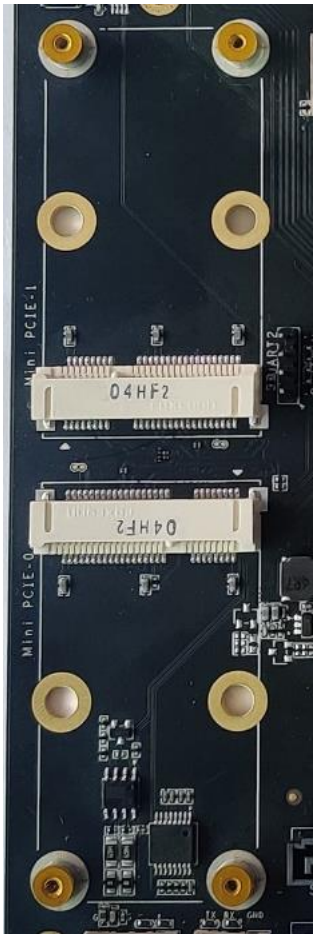

Figure 3-22 Mini PCIe3.0 Connector

3.16 SATA Interface

The DEMO board supports six SATA3.0 interfaces, in which, SATA1_TXP/TXN is directly output by controller, and the other SATA2_TXP/TXN is converted to 5 SATA interfaces output through SATA HUB chip JMB575

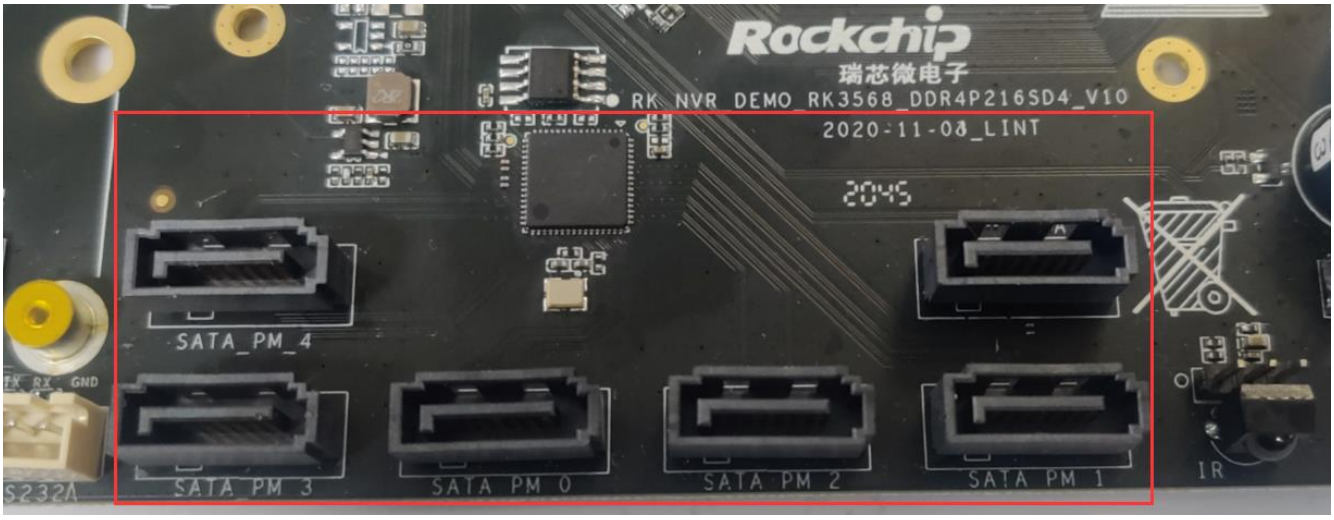

Figure 3-23 SATA Interface

3.17 Indicator Lights

There are three power indicator lights on the DEMO board, they are the working status indicator light, the Ethernet working indicator light, and the hard disk working indicator light from left to right respectively.

Figure 3-24 Indicator Lights

4 Notice

4.1 Notice

RK3568 DEMO board is suitable for lab or project environment. Please read the following notices before operation:

- ✧ Before unpacking and installing the DEMO board, please take the necessary anti-static measures to avoid the damage to the hardware of the DEMO board caused by ESD;
- ✧ Please hold the edge of the DEMO board, and do not touch the exposed metal part of the DEMO board so as to avoid the electrostatic damage to the components of the DEMO board;
- ✧ Please place the DEMO board on a dry plane to keep them away from heat source, electromagnetic interference source and radiation source, electromagnetic radiation sensitive equipment (such as medical equipment) and so on.

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